

Brenda Nogueira

 |  |  |  bcruznog@nd.edu

EDUCATION

University of Notre Dame

PhD in Computer Science and Engineering

Advisors: [Prof. Nitesh V. Chawla](#) & [Prof. Nuno Moniz](#)

Jan 2024 - Dec 2028

Notre Dame, IN, United States

University of Porto

Master in Data Science

Advisors: [Prof. Nuno Moniz](#) & [Prof. Rita Ribeiro](#)

Sep 2022 - Dec 2024

Porto, Portugal

University of Porto

Bachelor in Mathematics with Minor in Computer Science

Sep 2017 - Jun 2022

Porto, Portugal

Instituto Federal do Rio Grande do Sul

Technician in Administration Integrated to High School

April 2013 - Dec 2016

Canoas, Brazil

PUBLICATIONS [[GOOGLE SCHOLAR](#)]

Preprints and Under Submissions

Rethinking Evaluation in Compound Potency Prediction

Brenda Nogueira, Nuno Moniz, Connor W. Coley, Nitesh Chawla.

Machine Learning. Springer. 2026

Peer-Reviewed Publications

SPECTRA: Spectral Domain-Aware Graph Generation for Imbalanced Molecular Property Regression

Brenda Nogueira, Gisela Abigail Gonzalez-Montiel, Meng Jiang, Nitesh V Chawla, Nuno Moniz.

32nd ACM Special Interest Group on Knowledge Discovery and Data Mining Conference (KDD). 2026

Organic Chemistry as a Catalyst for AI Innovation: Challenges, Methods, and Emerging Paradigms

Nitesh V. Chawla, Gisela A. Gonzalez-Montiel,..., Nuno Moniz, [Brenda Nogueira](#),..., Olaf Wiest

Chemical Reviews. 2026

From Verification Burden to Trusted Collaboration: Design Goals for LLM-Assisted Literature Reviews

Brenda Nogueira, Werner Geyer, Andrew Andereson, Toby Li, John Kim, Nuno Moniz, Nitesh V. Chawla.

Association for the Advancement of Artificial Intelligence (AAAI). AI for Scientific Research Workshop. 2026

30st Annual ACM Conference on Intelligent User Interfaces (IUI). Poster Track. 2026

Spectral Manifold Harmonization for Graph Imbalanced Regression

Brenda Nogueira, Gabe Gomes, Meng Jiang, Nitesh V Chawla, Nuno Moniz.

42nd International Conference on Machine Learning (ICLR). DIG-BUGS Workshop

31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD). MLoG-GenAI Workshop.

2025

Automated Fish Size Measurement System for Long-Term Growth Studies in the Azores

Brenda Nogueira, Rita P. Ribeiro, Gui M. Menezes, Nuno Moniz

European Conference on Machine Learning and Knowledge Discovery (ECML/PKDD). SoGood Workshop.

2025

Graph-Imbalanced Regression for Rare Phenotypes

Brenda Nogueira, Nuno Moniz, Nitesh V Chawla.

12th Iberian Conference on Pattern Recognition and Image Analysis (IbPRIA). 2025

Experiential-Informed Data Reconstruction for Fishery Sustainability and Policies in the Azores

Brenda Nogueira, Gui M. Menezes, Nuno Moniz, Rita P. Ribeiro

Discover Data, Springer. 2025

Dynamics of Fisheries in the Azores Islands: A Network Analysis Approach

Brenda Nogueira, Ana Torres, Nuno Moniz, Gui M. Menezes.

22nd Portuguese Conference on Artificial Intelligence (EPIA). 2024

HONORS AND AWARDS

NSF Center for Computed Assisted Synthesis Travel Award

2026

Scientific Artificial Intelligence Graduate Fellowship

2025

NSF Center for Computed Assisted Synthesis Travel Award

2025

International Mathematics Without Borders Olympiad Brazilian Section | Gold medal

2016

EXPERIENCE

Guest Researcher <i>Los Alamos National Laboratory</i> Continuation of the Summer Internship in Advancing Machine Learning for Scientific Discovery	<i>Sept 2026 - Aug 2027</i>
Research Assistant <i>Lucy Institute for Data & Society</i> Exploration of Artificial Intelligence with a focus on imbalanced domains and chemistry data.	<i>Jan 2024 - Dec 2028 (Expected)</i>
Summer Internship <i>Los Alamos National Laboratory</i> Summer Internship in Advancing Machine Learning for Scientific Discovery	<i>Jun 2026 - Aug 2026</i>
Research Fellow <i>INESCTEC</i> Multi-objective optimization of raw material utilization for cutting and packing solutions in industry.	<i>July 2023 - Dec 2023</i>
Research Fellow <i>FCUP</i> Automated email-entity association pipeline using NLP.	<i>Oct 2022 - June 2023</i>
Research Initiation Fellow <i>INESCTEC</i> Machine learning for imbalance time series prediction about landings in the OKEANOS dataset.	<i>Aug 2022 - Sept 2022</i>
Intern <i>INESCTEC</i> Development of an online platform using Shiny/R for an AutoML solution.	<i>March 2022 - June 2022</i>
Intern <i>Ideavity</i> Development of a sales platform and applications using Outsystems.	<i>March 2022 - June 2022</i>
Intern <i>Evolutio</i> Website creation using wagtail (based on Django).	<i>Oct 2021 - Feb 2022</i>
Volunteer Intern <i>INESCTEC</i> Implementation of real-time face-based sleepiness recognition web-app.	<i>July 2021</i>

RESEARCH INTERESTS

Imbalanced Regression Learning: Designing data-driven solutions to enhance predictive performance in imbalanced and rare-event regression problems.

AI for Science: Applying AI and machine learning to real-world challenges in science and mathematics, including chemistry, biochemistry, and genomics—such as mathematical modeling, molecular design, virtual screening, and drug discovery.

Graph, Generative & Agentic AI: Developing graph-based, generative (VAE, diffusion), and agentic AI systems for synthetic data generation, predictive modeling, and autonomous decision-making.

PROFESSIONAL DEVELOPMENT / ACADEMIC ACTIVITIES

Graduate Student Cultural, University of Notre Dame, Kylemore Abbey, Ireland	2026
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ORGANIZER

Learning on Graphs Conference (LoG) Meetup	2025
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MENTORSHIP

John Kim , B.E. in Computer Science & Applied Math, University of Notre Dame	2025-Present
C-CAS GUIDE Program	2025-Present

SKILLS

Programming Languages and Technologies: Python, CUDA, PyTorch, TensorFlow, Deep Learning Frameworks, Scikit-Learn, R, Java.

Areas of Knowledge: Large Language Models, Synthetic Data Generation, Computer Vision, Partial Differential Equations (PDEs), Graph Learning, Contrastive Learning, Transformers, Time-Series, Generative Models, Statistical Analysis.

Web Technologies: HTML, CSS, JavaScript.